
ISO/IEC 27001 Compliance Baseline & Information Security Risk Management Report

Comprehensive Internal Audit, Evidence Mapping, and Core Asset Risk Treatment Pipeline

Target Organization: PayGuard Inc.

Scope: Production Infrastructure, Core FinTech Pipelines & Operations

Auditor Accountable: GRC Function & Audit Engineering Lead

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1. Executive Summary & Scope

During the second quarter of 2026, PayGuard Inc. initiated a full-lifecycle internal readiness audit to evaluate its information security environment against the **ISO/IEC 27001:2022** standard. The fundamental goal of this program was to establish an ironclad security posture across our core production systems, application layers, infrastructure pipelines, and company policies prior to engaging formal external certification bodies.

The audit scope explicitly encapsulates the cloud-native infrastructure hosted on Amazon Web Services (AWS), our fintech customer payment processing architecture, employee endpoint management configurations, and the corporate governance landscape. To facilitate an efficient audit and ensure long-term visibility for internal leadership, a centralized **Executive Compliance Dashboard** was constructed in Jira, tracking 99 distinct control elements across four key organizational domains.

Key Assessment Metric: Out of 99 total control tracking items, 91 controls have been validated as **Fully Compliant** via centralized macro-evidence mapping. 4 critical infrastructure gaps were identified within the Technological domain, which are currently isolated and under active technical remediation with strict SLAs.

2. Dashboard Architecture & Control Mapping

To prevent compliance fragmentation, all controls from standard Annex A groups were consolidated into a unified Jira ledger and categorized using structural metadata labels. This setup enables automated, real-time executive dashboard reporting grouped by status and control domain, providing a permanent data lineage from raw technical evidence up to board-level reporting.

Control Domain Allocation Breakdown

- **Organizational-Control (38 Total):** Governance rules, strategic policies, risk assessments, and vendor relationship frameworks.
- **Technological-Control (38 Total):** Network architecture, access control, identity lifecycle, cryptographic setups, and logging systems.
- **Physical-Control (14 Total):** Physical perimeter boundaries, asset protection, and structural resilience factors.
- **People-Control (8 Total):** HR management, onboarding validation, background screening, and strict security exit parameters.

3. Internal Audit Execution & Technical Deep-Dive

While administrative and physical domains were validated using macro-evidence aggregation, the **Technological-Control** domain was subjected to targeted, fine-toothed engineering checks. Below are the definitive testing procedures, evidence reviews, and compliance evaluations that were executed for the core technological infrastructure:

A. Control A.8.24 — Use of Cryptography

Testing Objective: Verify that cryptographic keys are generated, utilized, and rotated automatically to safeguard production tables and active file systems.

Evidence & Identified Gap: DevOps submitted cloud templates confirming production RDS PostgreSQL instances are fully encrypted at rest. However, evaluation revealed the environment relied on the default AWS-managed KMS Key instead of a Customer Managed Key (CMK), preventing fine-grained IAM policy control and omitting automatic annual key rotation.

Status: **PARTIALLY COMPLIANT** | **Action Item:** Migrate database to an explicit Customer Managed Key with automated 365-day rotation policies enabled.

B. Control A.8.12 — Data Leakage Prevention

Testing Objective: Ensure automated mechanisms prevent secret tokens, private keys, or payment details from leaking out via corporate tools or codebases.

Evidence & Identified Gap: Security engineering confirmed AWS Macie monitors public S3 buckets securely. However, GitHub Advanced Security native secret scanning was found active **exclusively on public repositories** due to license tier budgeting constraints, leaving internal private code repositories completely unmonitored for hardcoded strings or API tokens.

Status: **PARTIALLY COMPLIANT** | **Action Item:** Upgrade GitHub licensing structures and enable automated secret scanning across all private internal code repositories.

C. Control A.8.20 — Network Security

Testing Objective: Verify logical network isolation barriers exist to shield production payment pathways from external environments.

Evidence & Identified Gap: Network routing tables confirmed isolated AWS accounts for environments. However, a live, bi-directional AWS VPC Peering connection was discovered actively linking the Staging VPC to the Production VPC to facilitate rapid developer testing, presenting an extreme lateral movement pivot risk.

Status: **FINDING IDENTIFIED** | **Action Item:** Decommission the cross-environment VPC Peering connection immediately and deploy localized API mocking frameworks.

D. Control A.8.16 — Monitoring Activities

Testing Objective: Audit centralized SIEM log collection pipelines and ensure administrative activities generate real-time alerting anomalies.

Evidence & Identified Gap: System logs successfully route to a centralized Datadog tenant. However, there were zero real-time automated alerts configured for critical cloud control plane changes (e.g., IAM modification or root access), with reviews handled manually on a bi-weekly cycle. Additionally, administrative log mutation guardrails were absent.

Status: **PARTIALLY COMPLIANT** | **Action Item:** Deploy automated Datadog monitors for structural IAM updates and enforce rigid S3 WORM / Object Lock compliance for CloudTrail buckets.

4. Streamlined Processing via Bulk Evidence Mapping

To prevent severe project exhaustion and maintain operational speed, the remaining 89 controls within the Jira backlog were handled concurrently through **Macro-Evidence Packages**. Instead of analyzing controls one-by-one via individual interviews, the GRC function aggregated centralized systemic artifacts that comprehensively addressed broad regulatory domains simultaneously:

Evidence Package Name	Core Internal Artifacts Reviewed	ISO 27001 Controls Satisfied
Corporate Policy Handbook	2026 InfoSec Master Policy, Access Control Policy, Asset Management Policy, Approved SDLC Guidelines.	~30 Controls under Organizational-Control (A.5.* groups)
Third-Party Vendor Audit Package	Annual Amazon Web Services (AWS) SOC 2 Type II Compliance Audit Report, verified data-center physical ingress logs.	~20 Controls under Physical-Control (A.7.* groups)
HR & Personnel Records Package	Signed non-disclosure agreements (NDAs), standard background verification checks, employee onboarding checklist logs.	~15 Controls under People-Control (A.6.* groups)
IT Endpoint Operational Fleet Report	MDM configuration sheets (Kandji/Jamf) verifying 100% corporate fleet patch compliance, centralized laptop backup logs.	~20 Controls under Technological-Control (A.8.1, A.8.14)

Using Jira's native bulk operations engine, all controls matching these criteria were moved to a status of **DONE** with a standardized verification comment attached to preserve the compliance trail. This streamlined approach ensured that administrative overhead was minimized, allowing engineering resources to remain hyper-focused on fixing the four active technological vulnerabilities discovered.

5. Information Security Risk Register

Pursuant to the mandatory risk assessment requirements detailed within ****ISO/IEC 27001 Clause 6.1.2****, all identified technical security gaps have been formalized into a quantitative corporate Risk Register. The Inherent Risk Rating is mapped directly to a classical qualitative matrix derived through the mathematical calculation:

$$\text{Risk Score} = \text{Inherent Likelihood} \times \text{Inherent Impact}$$

Scores range from 1 to 25, where any rating exceeding 12 demands immediate corrective remediation, board visibility, and active engineering assignment.

Risk ID	Threat Scenario Description	Like (1-5)	Imp (1-5)	Inherent Rating	Treatment Policy	Linked Jira Ticket
RSK-01	Lateral Movement via Network: A compromise of a staging server lets an attacker bridge into production databases via active VPC peering.	3	5	15 (HIGH)	Mitigate: Terminate VPC route immediately.	`KAN-168` IN PROGRESS
RSK-02	Credential Exposure: Hardcoded cloud keys or DB tokens committed to internal private repos are exploited by an adversary.	4	4	16 (HIGH)	Mitigate: Enable private repo Git secret scanning.	`KAN-169` IN PROGRESS
RSK-03	Data Breach via Backup: Database backups are accessed by unauthorized personnel due to loose default AWS-managed KMS IAM policies.	2	5	10 (MED)	Mitigate: Migrate to explicit CMKs with tight policies.	`KAN-27` IN PROGRESS
RSK-04	Delayed Incident Detection: Malicious AWS control plane drift or exfiltration goes undetected due to lack of real-time SIEM alerts.	3	4	12 (MED)	Mitigate: Deploy automated alerts and WORM storage.	`KAN-15` IN PROGRESS

6. Next Steps & Project Governance

With the compliance architecture firmly established, governance will proceed according to the following cadence:

1. **SLA Monitoring:** High-risk technical tickets (`RSK-01` and `RSK-02`) carry a strict 14-day SLA, requiring resolution by July 7, 2026. Medium-risk tickets carry a 30-day window.
2. **Residual Risk Validation:** Once engineering updates are committed, the GRC function will perform formal regression testing to close out the Jira tickets and re-evaluate the Residual Risk profile down to Low (\$\le 4\$).
3. **External Stage 1 Audit:** Upon reaching 100% dashboard completion, the compiled Jira ledger will be locked and exported as definitive evidence for third-party registrars.